

# Position Generalization in Translated FashionMNIST

Architecture, patch scale, and patch projection

Experimental report

6 configurations / 12 fits / 24 evaluations

*Abstract.* Six configurations were fitted on random or centered object positions. Across all six, the absolute A -> B change is at most 0.90 percentage points, whereas B -> A drops by 57.74 to 75.58 points. CNN leads all settings (35.40% on B -> A). Patch size 8 leads three ViT settings; the two patch projections differ by at most 0.61 points.

## 1 Study question

A classifier can attach class evidence to location. To isolate this effect, each 28 x 28 FashionMNIST image in A is assigned one deterministic position sampled from the 1,369 valid integer locations; B always uses the centered position. Placements are fixed per sample and split for a given seed. The experiment asks which model choices retain accuracy when the test position distribution changes.

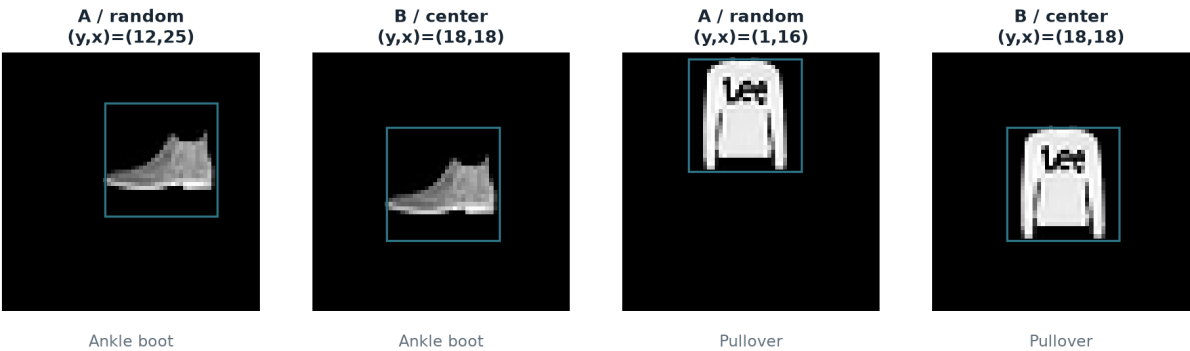


Figure 1. Two official FashionMNIST items under A and B. The teal outline marks the 28 x 28 source support; only its canvas position changes.

| POSITION SUPPORT         | TRAINED MODELS         | FINAL TESTS                     |
|--------------------------|------------------------|---------------------------------|
| 1,369 vs 1               | 12                     | 24                              |
| A locations / B location | 6 configurations x A/B | four settings per configuration |

### 1.1 Questions tested

| COMPARISON 1                | COMPARISON 2                     | COMPARISON 3                    |
|-----------------------------|----------------------------------|---------------------------------|
| Architecture                | Patch scale                      | Projection                      |
| Which model handles B -> A? | How does token count affect ViT? | Do equivalent maps train alike? |

# Controlled Protocol

One split, one training budget, and four evaluation settings

## 2 Evaluation matrix

| Setting | Training | Testing | Purpose                              |
|---------|----------|---------|--------------------------------------|
| A -> A  | random   | random  | in-distribution reference            |
| B -> B  | center   | center  | fixed-position reference             |
| A -> B  | random   | center  | transfer within A's position support |
| B -> A  | center   | random  | one-to-many position transfer        |

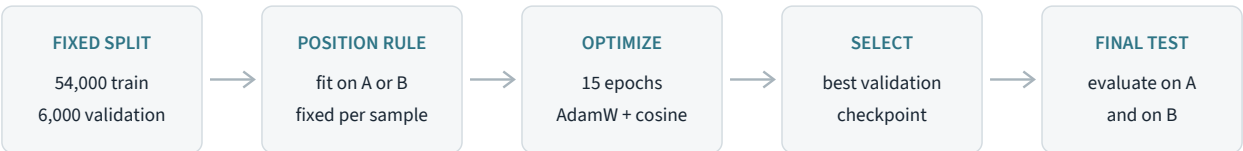
Table 1. The official test partition is used only after checkpoint selection.

## 3 Controlled comparisons

| Group        | Configurations           | Changed factor / fixed factors                                                       |
|--------------|--------------------------|--------------------------------------------------------------------------------------|
| Architecture | MLP, CNN, ViT p=8        | Architecture changes; split, optimizer, budget, and evaluation stay fixed.           |
| Patch scale  | ViT p=4, 8, 16           | Token grid changes; Transformer width, depth, heads, and projection type stay fixed. |
| Projection   | Conv2d, Flatten + Linear | Patch projection changes at p=16; the Transformer body is identical.                 |

Table 2. Each group changes one stated design choice; the architecture group is not parameter-matched.

## 4 Run sequence



One configuration: two fits and four held-out test measurements

Figure 2. Every fit uses a validation checkpoint, then both held-out test distributions are evaluated with shared deterministic placements.

**Direction matters.** B -> A extrapolates from one observed location to 1,369 possible locations. A -> B evaluates a location already contained in A's position support.

# Implementations

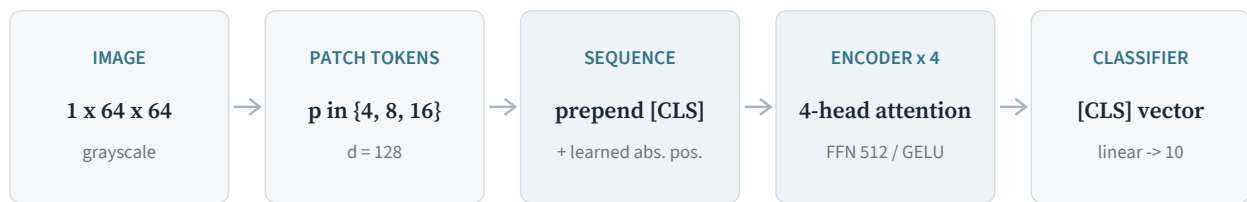
Model size, positional representation, and optimization settings

## 5 Model definitions

| Model | Core structure                                                                              | Spatial treatment                   | Parameters                              |
|-------|---------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------|
| MLP   | 4096 -> 128 -> 128 -> 10; LayerNorm; GELU; dropout 0.1                                      | absolute input pixels               | 542,474                                 |
| CNN   | channels 1-32-32-64-64-128; BatchNorm; two max-pools; adaptive 2 x 2 pool; 512 -> 128 -> 10 | shared local filters                | 205,994                                 |
| ViT   | d=128; depth 4; heads 4; FFN 512; pre-norm; GELU; dropout 0.1                               | learned absolute position embedding | p4 829,834<br>p8 811,402<br>p16 829,834 |

Table 3. Trainable parameter counts are taken from the fitted models.

## 6 Vision Transformer



Pre-norm Transformer; dropout 0.1; non-overlapping patches

Figure 3. ViT implementation shared by the patch-scale and projection studies.

**Position representation.** The ViT adds one learned absolute embedding to each patch index. Translation equivariance is therefore not built into the token sequence.

## 7 Training recipe

| Item      | Controlled choice                                                        |
|-----------|--------------------------------------------------------------------------|
| Data      | 28 x 28 FashionMNIST image on a 64 x 64 black canvas                     |
| Split     | 54,000 train, 6,000 validation; official 10,000-image test set held out  |
| Training  | 15 epochs; batch 64; cross-entropy; AdamW, lr 0.001, weight decay 0.0001 |
| Selection | cosine annealing; highest validation accuracy checkpoint                 |
| Reporting | top-1 test accuracy; elapsed training time; seed 42                      |

Table 4. Formal environment: PyTorch 2.11.0+cu128; CUDA 12.8; NVIDIA GeForce RTX 5070 Laptop GPU; Linux (WSL2). Mixed precision enabled.

# Learning Curves and Checkpoints

Validation behavior for MLP, CNN, and patch-8 ViT over 15 epochs

8 Training dynamics

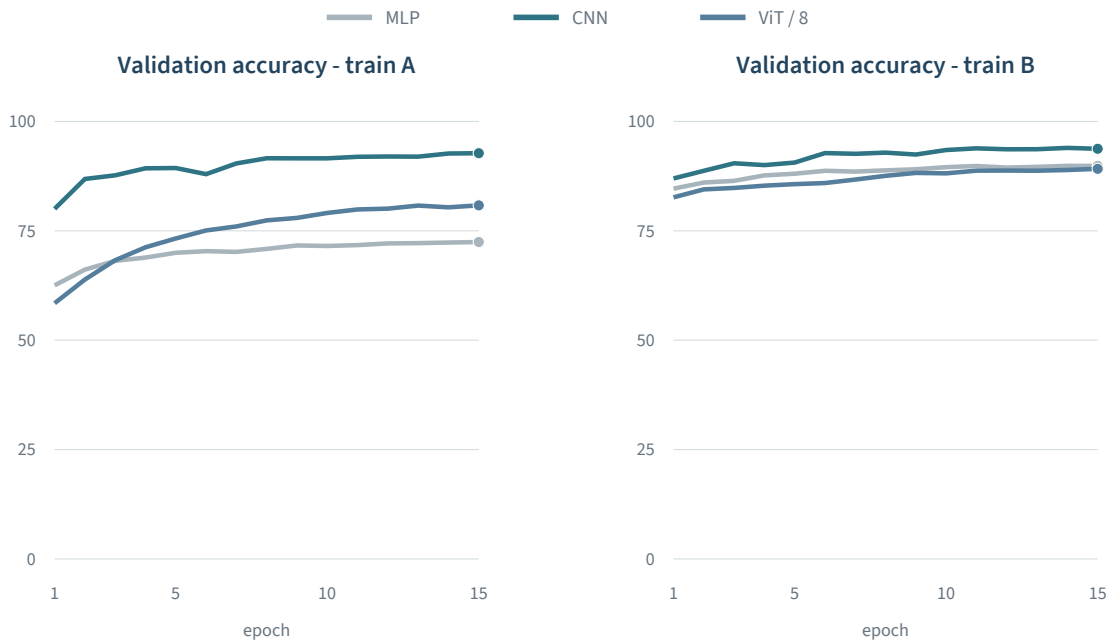


Figure 4. Validation accuracy for A-trained and B-trained architecture models.

| Configuration | Train A           | Train B           |
|---------------|-------------------|-------------------|
| MLP           | 72.43% (epoch 15) | 89.85% (epoch 14) |
| CNN           | 92.73% (epoch 15) | 93.95% (epoch 14) |
| ViT, patch 8  | 80.82% (epoch 15) | 89.17% (epoch 15) |

Table 5. Best validation accuracy and selected checkpoint epoch.

**Checkpoint selection.** The six checkpoints occur at epoch 14 or 15. CNN gives the highest validation accuracy for both training distributions; each model validates higher on centered images.

| SELECTED EPOCHS           | BEST VALIDATION | LARGEST A/B GAP |
|---------------------------|-----------------|-----------------|
| 14-15                     | 93.95%          | 17.42 pp        |
| all six architecture fits | CNN / train B   | MLP validation  |

# MLP, CNN, and ViT

Shared data split and optimizer; model capacity is not matched

9 Test accuracy

| Configuration | Parameters | A -> A | B -> B | A -> B | B -> A |
|---------------|------------|--------|--------|--------|--------|
| MLP           | 542,474    | 71.89  | 89.57  | 71.97  | 13.99  |
| CNN           | 205,994    | 92.35  | 93.14  | 92.87  | 35.40  |
| ViT, patch 8  | 811,402    | 80.40  | 88.27  | 81.13  | 18.81  |

Table 6. Top-1 test accuracy (%); teal values mark column leaders.

Accuracy on fixed-center train -> random-position test

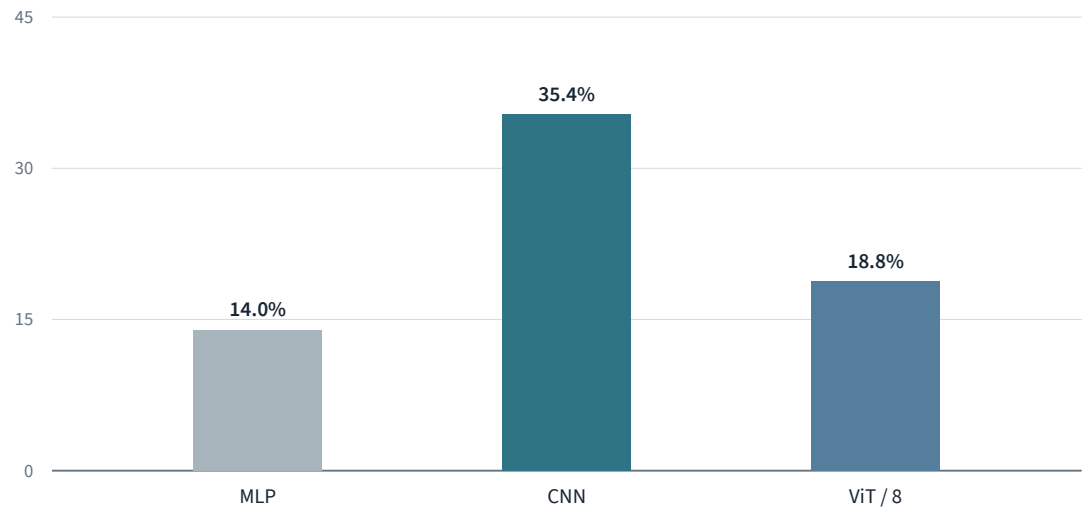


Figure 5. CNN reaches 35.40% on B -> A, 16.59 points above patch-8 ViT.

|                                                 |                                                                              |                                                                          |
|-------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>B -&gt; A LEADER</b><br><b>35.40%</b><br>CNN | <b>SMALLEST B -&gt; A DROP</b><br><b>57.74 pp</b><br>CNN, relative to B -> B | <b>A -&gt; B CHANGE</b><br><b>0.73 pp max</b><br>all three architectures |
|-------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------|

9.1 Interpretation

CNN leads all four settings despite having fewer parameters than the MLP and ViT. Shared local filters are a plausible reason for the stronger shift result, but this run does not separate architecture from parameter count or model-specific hyperparameter tuning.

# The Transfer Asymmetry

Accuracy changes when the training and test position supports differ

## 10 Directional shift

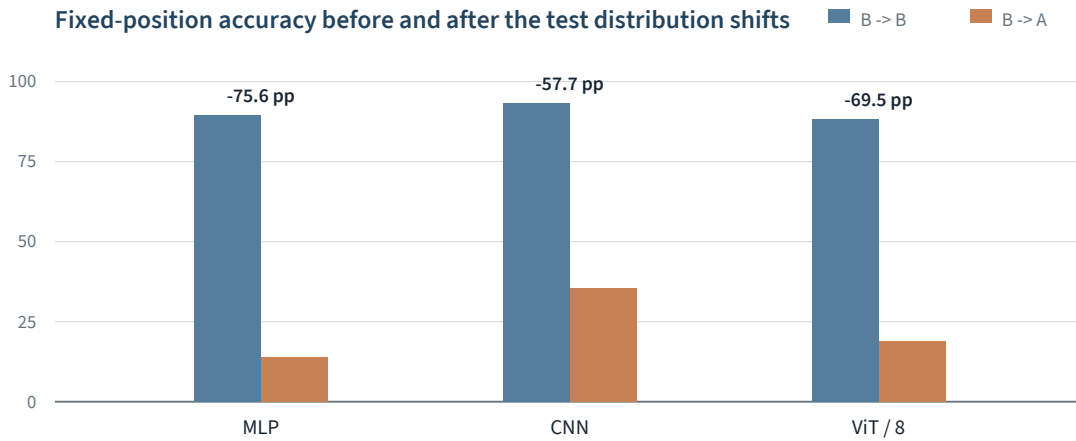


Figure 6. Centered-only training drops by 57.74 to 75.58 points under random-position testing.

| Configuration   | A -> B minus A -> A | B -> A minus B -> B |
|-----------------|---------------------|---------------------|
| MLP             | +0.08 pp            | -75.58 pp           |
| CNN             | +0.52 pp            | -57.74 pp           |
| ViT, patch 4    | -0.31 pp            | -70.20 pp           |
| ViT, patch 8    | +0.73 pp            | -69.46 pp           |
| ViT, patch 16   | +0.90 pp            | -71.96 pp           |
| ViT, p16 Linear | +0.81 pp            | -72.50 pp           |

Table 7. Signed test-distribution change for all six configurations.

**Consistent pattern.** After training on A, moving the test objects to the center produces an absolute change of at most 0.90 percentage points. After training on B, random placement removes at least 57.74 points.

The comparison supports a coverage explanation: A exposes the model to the center and many off-center positions, whereas B provides no direct evidence about most of A. The result does not imply that A is universally harder; it shows that high B -> B accuracy is not evidence of translation robustness.

# Patch Scale

ViT body fixed at d=128, depth 4, four heads, and FFN width 512

## 11 Patch-size results

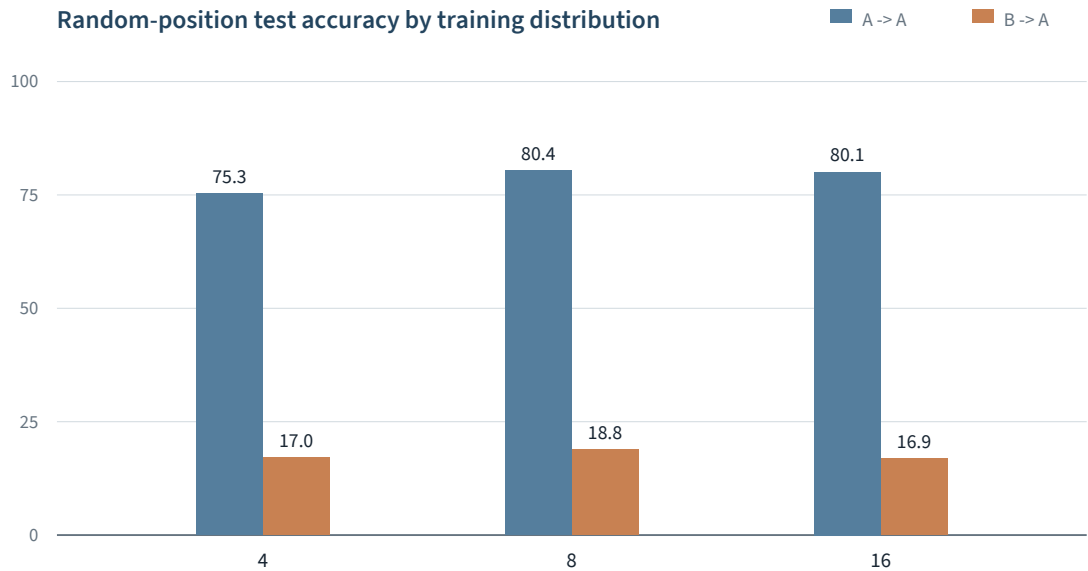


Figure 7. Selected random-position outcomes; Table 8 reports all four settings.

| Patch | Tokens | A -> A | B -> B | A -> B | B -> A | Time     |
|-------|--------|--------|--------|--------|--------|----------|
| 4     | 256    | 75.28  | 87.22  | 74.97  | 17.02  | 8.93 min |
| 8     | 64     | 80.40  | 88.27  | 81.13  | 18.81  | 7.36 min |
| 16    | 16     | 80.07  | 88.87  | 80.97  | 16.91  | 6.46 min |

Table 8. Accuracy (%) and summed training time; teal marks the best value in each column.

**Observed trade-off.** Patch size 8 leads on A -> A, A -> B, and B -> A. Patch size 4 expands the sequence to 256 tokens and takes longer without improving accuracy under the shared 15-epoch budget.

Patch size 16 is fastest and gives the best B -> B result. Under this protocol, patch size 8 records the highest A -> A, A -> B, and B -> A accuracy with 64 tokens. Runtime is descriptive because it depends on hardware and execution order.

| PATCH 4                 | PATCH 8                 | PATCH 16                |
|-------------------------|-------------------------|-------------------------|
| 65,536                  | 4,096                   | 256                     |
| patch-patch pairs (N^2) | patch-patch pairs (N^2) | patch-patch pairs (N^2) |

# Patch Projection

Conv2d and Flatten + Linear at patch size 16

## 12 Equivalent projections

**CONV2D PROJECTION**

**kernel = stride = 16**

1 x 16 x 16 -> 128 channels

**FLATTEN + LINEAR**

**unfold into 16 x 16 patches**

256-vector -> shared Linear(256,128)

Both return 16 tokens x 128 dimensions and represent the same affine map

Figure 8. With non-overlapping patches, the two implementations differ in layout, not in the affine function they can represent.

| Test   | Conv2d | Flatten + Linear | Absolute gap |
|--------|--------|------------------|--------------|
| A -> A | 80.07  | 79.67            | 0.40         |
| B -> B | 88.87  | 88.80            | 0.07         |
| A -> B | 80.97  | 80.48            | 0.49         |
| B -> A | 16.91  | 16.30            | 0.61         |

Table 9. Test accuracy (%) at p=16; maximum absolute gap 0.61 points.

**Result.** The trained projections remain numerically close. Because the record contains one seed, the residual gap is not treated as evidence that either implementation is better.

## 13 Conclusions and limits

- 01**

**Transfer is strongly asymmetric.** For A-trained models, the absolute change at center is at most 0.90 percentage points. B-trained models drop by 57.74 to 75.58 points at random positions. CNN leads all four settings.
- 02**

**Patch size 8 leads three of four settings.** It gives the best B -> A result without the 256-token cost of patch size 4.
- 03**

**Equivalent patch projections stay close in this run.** Conv2d and Flatten + Linear differ by at most 0.61 percentage points.

| Scope                                                                                                                                                                                                                                                                                  | Interpretation rule                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>- One training and placement seed; no variance estimate.</li><li>- Model sizes and optimal hyperparameters are not matched.</li><li>- Only translation on a black canvas is tested.</li><li>- Runtime is hardware- and order-specific.</li></ul> | <p>Small differences are described only as numerically close in this run; no significance claim is made.</p> <p>Conclusions apply to the shared protocol rather than all possible training recipes.</p> |